# Software Requirements Specification

for

# P.A.T.A.N.G.

**Version 1.0**

**Prepared by**


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# Revisions

| Version | Primary Author(s) | Description of Version | Date Completed |
|---|---|---|---|
| 1.0 | Looney Tunes | Initial draft of SRS | 21/01/2026 |
| 1.1 | Looney Tunes | The draft was reviewed by TA and final changes were made. | 23/01/2026 |

# 1 Introduction

## 1.1 Product Scope

Our product aims to streamline the process for Students, Faculty members, and Campus members in general with respect to various facilities that are present on campus and helping in ensuring fair and maximum utilization of these resources. The major benefits are:

- **Gym and Swimming Slot Registrations:** Reducing the manual work of registering and paying in person, by digitizing the process of registration, payments, and allocation. Further ensuring access and fair use of the methods of QR codes.

- **Sports Facilities Booking:** Allowing members of the campus community to book sports facilities ahead of time to ensure no clashes, and fair use between people in the campus community, which in turn will make facilities like Badminton Courts, Squash Courts, Tennis Courts, Basketball Courts and Table Tennis Courts

- **New-SAC Facilities Booking:** Helping Gymkhana Executives, Club Coordinators, and other entities who use the shared facilities at New-SAC such as the Literary Society Room, Senate Hall, and Multi-purpose Room

- **Gymkhana Calendar:** Creating a unified calendar, with categorization to allow students to view the upcoming events on campus and any registration details that are required with it. Allowing students to go straight to the registration portals from this calendar. Gymkhana executives can also upload their events on this calendar and make it easier for the camps community to view all events at a single location.

- **Feedback:** A unified feedback portal which will take feedback for the software and the related Gymkhana/Campus facilities and pass the same to the concerned people to help them rectify the issue.

By centralizing these features, we aim to create a unified portal which helps the Students and Staff better manage and utilize campus resources, increasing the social welfare on campus.

## 1.2 Intended Audience and Document Overview

### 1.2.1 Document Overview

The document consists of the following sections:

- **Introduction** - Provision of basic terminologies and information used that will help the reader understand the document better. Can be skipped if the reader is familiar with the terminology.

- **Overall Description** - Offers precise and concise information regarding the functionality, interface, assumptions and dependencies of the software. Recommended to be viewed by all readers.

- **Specific Requirements** - Offers a detailed description of the software, its functionality, interfaces and working by dividing the entire user experience into multiple use cases.

- **Non-Functional Requirements** - Describes the performance, security and other qualities that are essential for the safety and smooth running of the software.

### 1.2.2 Intended Audience

- **Developers:** Developers will use this document to understand the functional and non-functional requirements, system architecture, access control logic, and technical constraints of the ***Patang*** system. They will focus on detailed feature requirements, user flows, and backend responsibilities such as slot booking restrictions, penalty handling, and user authentication logic.

- **Project Managers:** Project managers will use the document to track the project's scope, stakeholder expectations, and priorities. It provides a roadmap of deliverables and dependencies that help in resource planning and provide their insights on the functionalities of the projects.

- **Testers:** Engineers who will verify the defined use cases, workflows, and penalty conditions to design and execute the test case. Ensuring implemented system aligns with the criterias in this document.

- **End Users:** Faculty, Students, Gymkhana Executives, Administrative Staff, On-ground workers may refer to the overview of the project and the use-cases to understand the functionalities that are being implemented.

## 1.3 Definitions, Acronyms and Abbreviations

- SAC – Student Activity Center
- OTP- One Time Password
- QR- Quick Response
- ID Card- Identity Card

## 1.4 Document Conventions

*<In general this document follows the IEEE formatting requirements. Use Arial font size 11, or 12 throughout the document for text. Use italics for comments. Document text should be single spaced and maintain the 1" margins found in this template. For Section and Subsection titles please follow the template.*
*TO DO: Describe any standards or typographical conventions that were followed when writing this SRS, such as fonts or highlighting that have special significance. Sometimes, it is useful to divide this section to several sections, e.g., Formatting Conventions, Naming Conventions, etc.>*
The document follows the IEEE formatting requirements:

- The headings of all sections are written in **bold** using Arial font size 14
- The headings of subsections are **bold** and written in Arial font size 12
- The content of sections is written in Arial font size 11
- 1" margins are maintained
- Bullet point ordering has been used for lists
- Italics are used for comments
- Document text is single spaced

## 1.5 References and Acknowledgments

**References:**

Style Guide (for this document): IEEE Software Requirements Specification Template
Designing UI and Use Case Models: Moqups and Canva
Designing Flowcharts: Lucidchart

# 2 Overall Description

## 2.1 Product Overview

The proposed system aims to be a new self-contained resource management product designed to centralize and digitize the booking, allocation, monitoring and visibility of various campus facilities and activities/events. Currently, the campus relies on a combination of manual registrations, offline approvals, informal coordination, and disconnected tools for managing/booking sports facilities, common spaces in New-SAC, and event visibility. These fragmented processes often result in scheduling conflicts, inefficient utilization of facilities, lack of transparency, and increased administrative burden.

This product is conceived as a unified web-based portal that serves as a single point of interaction for students, Gymkhana executives, coaches and caretakers for all the student gymkhana activities. The system integrates sports facility booking/registration, New-SAC room booking and a centralized Gymkhana calendar under a common authentication and authorization framework. Each subsystem follows standardized booking rules, approval workflows, and fair-use policies while sharing common services such as user management, notifications and penalty handling.

The product operates as an independent service but interfaces with existing campus infrastructure such as the institute email system for OTP-based authentication. On-ground operations are supported through caretaker and administrator dashboards that enable real-time verification of bookings, attendance marking using roll numbers or QR codes, and release of unused slots. By enforcing policy-driven constraints and providing role-based access control, the system ensures fairness, accountability, and maximum utilization of campus resources.

## 2.2 Product Functionality

1. Customer Interface:

Customers are the primary consumers of campus facilities and services. The system enables them to access resources in a fair, transparent, and user-friendly manner.

Functionalities:

- Book sports facility slots (Badminton, Tennis, Squash, Basketball, Table Tennis, etc.) under defined fair-use constraints
- View real-time availability of sports facilities and booked slots
- Book slots up to 3 days in advance, limited to 2 slots in a rolling three-day window
- Use the “Look for Groups” feature to form groups for larger courts, improving utilization efficiency
- Automatically lose bookings if group formation is incomplete by the booking day
- Cancel bookings within stipulated time to avoid penalties
- Receive penalties (temporary booking suspension) for absence or misuse
- Apply for Gym and Swimming subscriptions (monthly/semesterly/yearly)
- Upload required documents and SBI Collect payment receipts
- Access gym/swimming facilities using QR-code based digital access pass
- View all ongoing and upcoming campus activities via the Unified Gymkhana Dashboard

- Filter events by category and access event details and registration links
- Submit feedback related to:
    - App functionality and bugs
    - Facility-related issues such as equipment, cleanliness, or staff behavior

2. Caretaker Interface

Caretakers are responsible for real-time facility supervision and operational enforcement at ground level.

Functionalities:

- View daily schedules and slot bookings for assigned facilities
- Verify user identity using Roll Number / ID Card / QR Code
- Mark users as Present or Absent based on timely arrival
- Automatically release unused slots into the system for walk-in users if a user is not present
- View facility-related feedback submitted by users
- Approve or reject gym and swimming registrations after document and payment verification
- Support fair-use enforcement by reporting misuse or irregularities

3. Coaches / Captains Interface

Coaches and team captains manage institute-level sports activities and require priority access for team practices.

Functionalities:

- Override standard booking limits for approved institute teams
- Block specific time slots to reserve facilities for official practices or training sessions
- View schedules and availability relevant to their teams
- Access and review feedback related to their respective sports facilities
- Coordinate with caretakers and Gymkhana executives for operational alignment

4. Gymkhana Executives Interface

Gymkhana Executives act as administrative authorities responsible for policy enforcement, approvals, and content moderation.

Functionalities:

- Approve or reject:
    - Sports facility overrides and event-based bookings
    - Event submissions for the Gymkhana calendar
- Upload and manage campus-wide events on the Unified Gymkhana Dashboard
- Delegate booking privileges to club coordinators for New-SAC facilities
- Manage New-SAC shared room allocation
- Block slots for institute events, maintenance, or large-scale activities
- View analytics related to facility usage and booking patterns
- Access all categories of feedback and route them to appropriate authorities

- Add or remove coordinators with specific access rights

## 2.3 Design and Implementation Constraints

Hardware Requirements:

- A functional computer or mobile device with a stable internet connection would be sufficient to access the application.
- Mobile devices must have a minimum of 2 GB of available RAM and an internet speed of at least 0.1 Mbps to ensure the application opens smoothly.

Software Requirements:

- The application is compatible with any devices and operating systems supporting modern
- browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, Internet Explorer and
- Safari and works better on latest versions of these browsers.

## 2.4 Assumptions and Dependencies

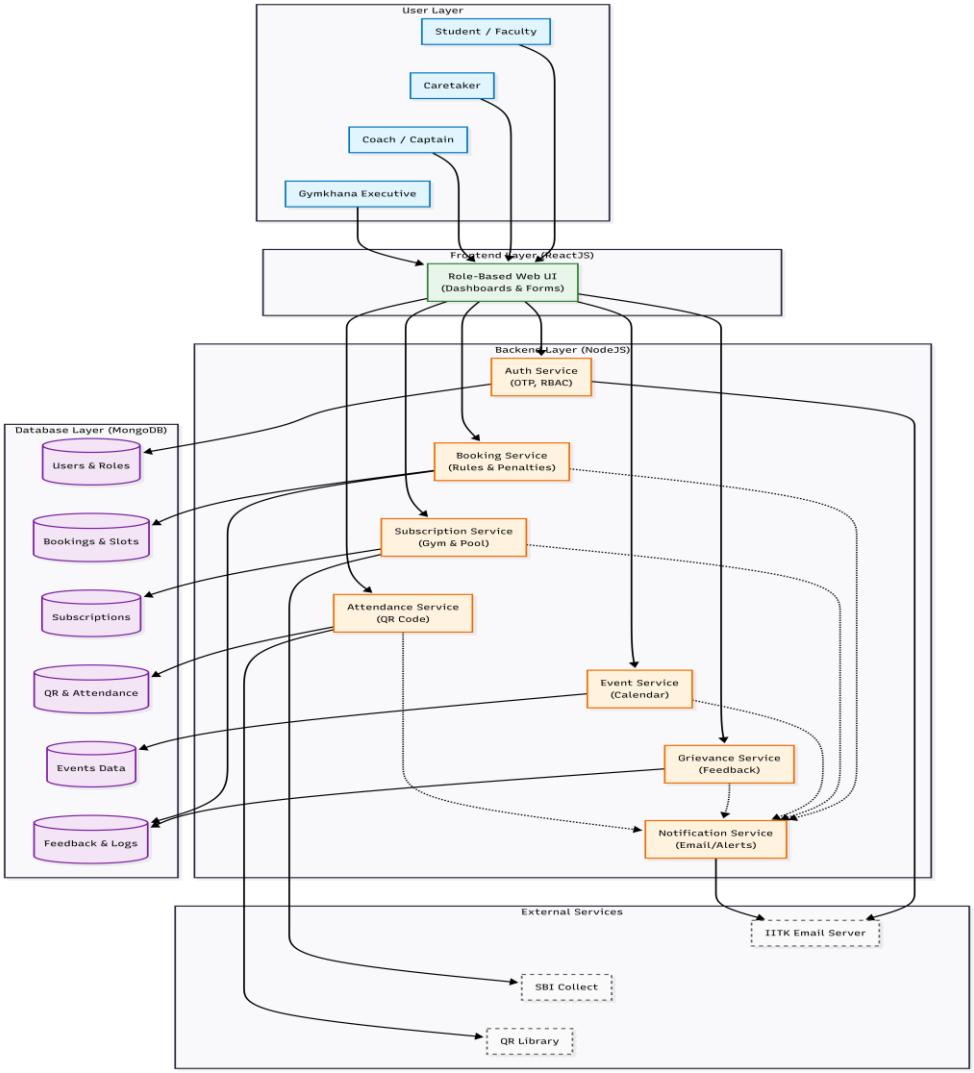
Assumptions

- The number of registered users on the platform will not exceed 10,000 active users (entire campus community including students, faculty members, caretakers, coaches etc.) at any point in time.
- The system is expected to handle a maximum of 500 concurrent users during peak booking hours.
- All users (students, faculty, caretakers, executives) possess a valid institute-issued email ID and have regular access to email for OTP-based authentication.
- Campus internet connectivity is sufficiently stable to support real-time booking, QR-code verification, and dashboard updates.
- Caretakers and administrators will actively use the system for attendance marking, slot verification, and operational updates.
- QR code scanning will be performed using smartphones or standard scanning devices available at facilities.
- Institute policies regarding slot allocation, penalties, and fair-use limits will remain consistent during the system's operational lifetime.
- Users are assumed to follow prescribed booking rules and system guidelines.

Dependencies

- Institute Email Service – Used for OTP-based user authentication and communication.
- Web Browser Compatibility – The system depends on modern web browsers for proper functionality.
- ReactJS – Frontend framework for building the web-based user interface.
- NodeJS- Backend Runtime Environment for server-side application logic.
- MongoDB- Database Management System for persistent data storage.
- Git and a Git-based repository hosting service for source code management.
- QR Code Libraries – Third-party libraries for QR code generation and scanning functionality.
- Cloud-based or institute-provided servers for deployment and availability.

## 2.5 System Architecture Diagram




# 3 Specific Requirements

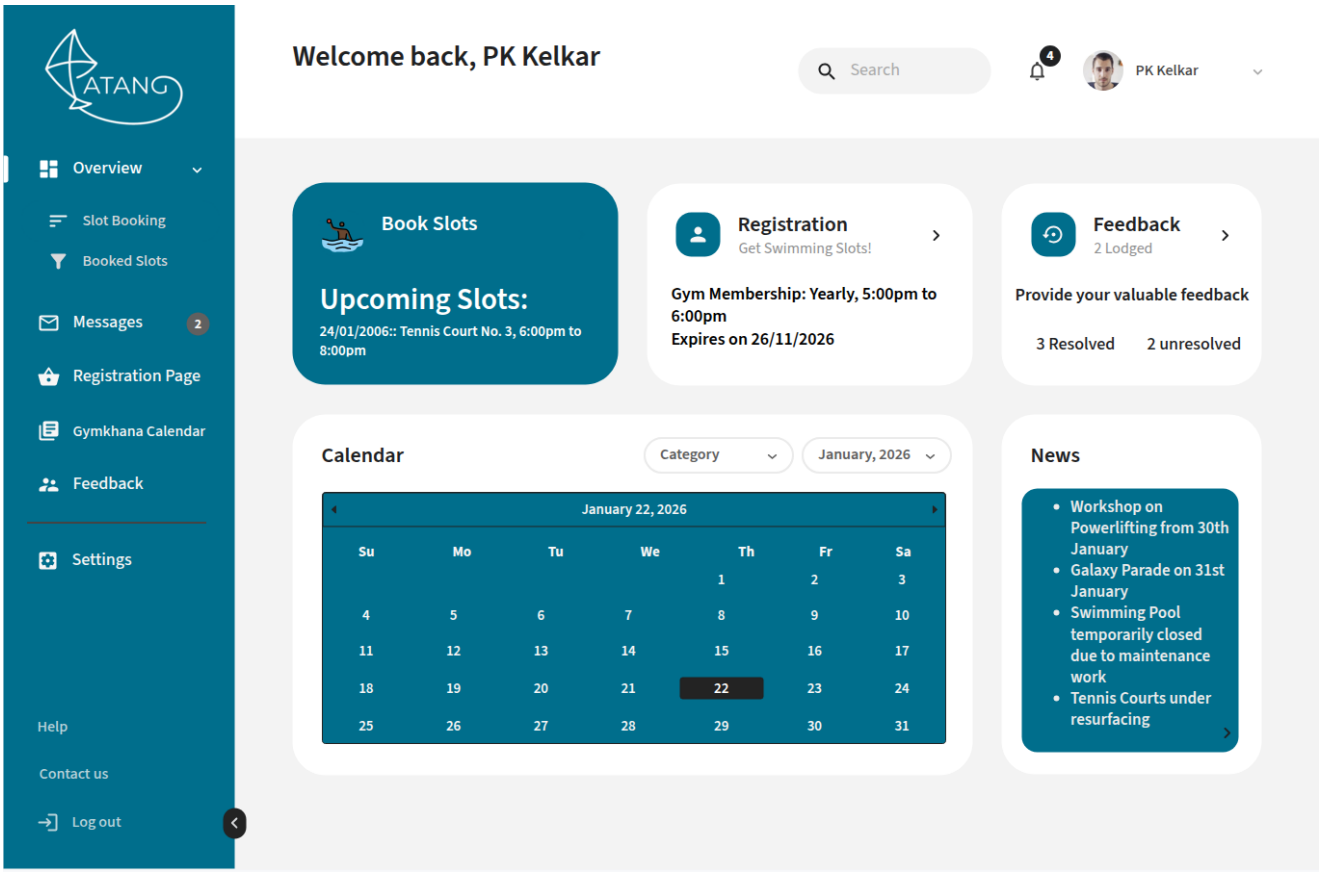
## 3.1 External Interface Requirements

### 3.1.1 User Interfaces

The application will be used by primarily:

- Customers: Students, Faculty Members, Other Campus community members
- Caretakers
- Coaches/Captains
- Gymkhana Executives

## Customers:



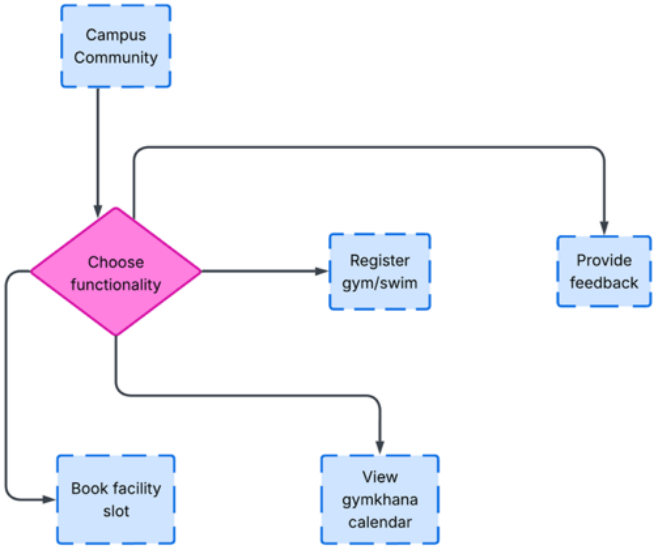
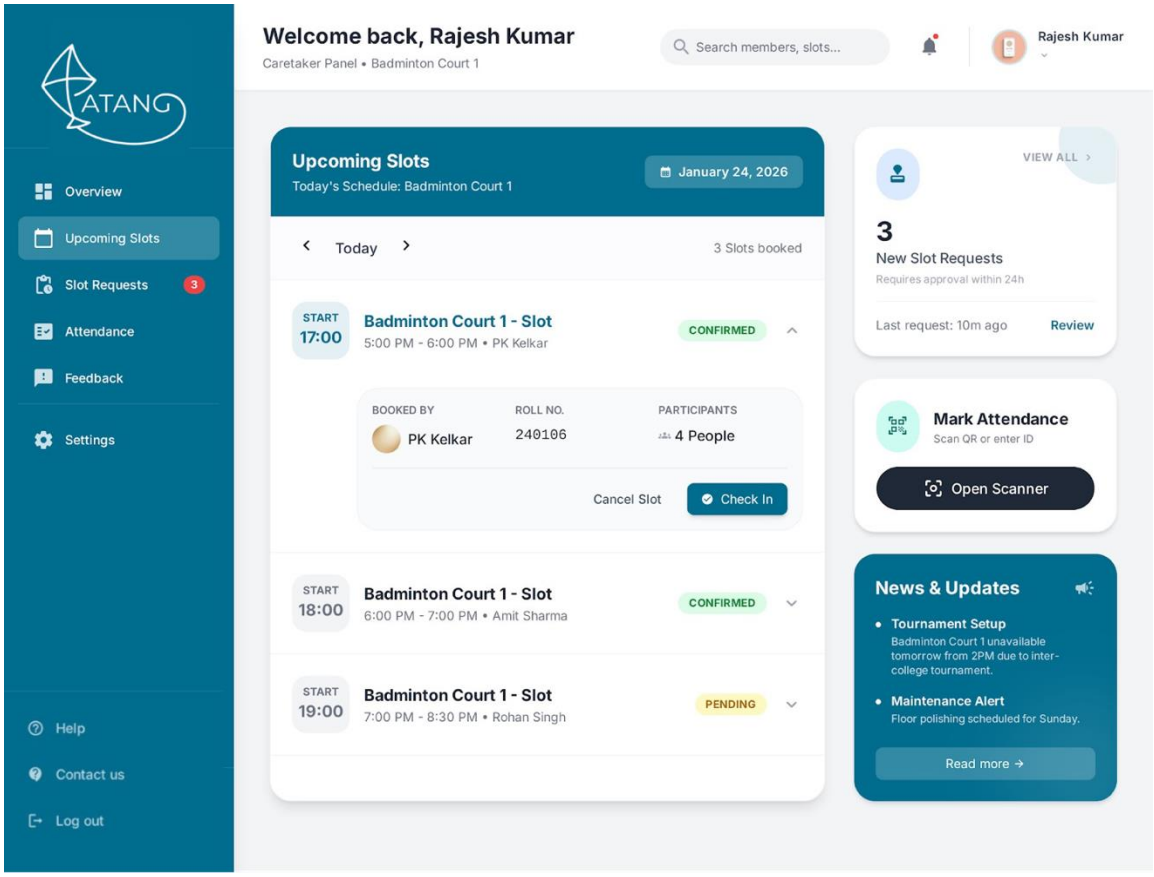




*Figure 1: Customer UI and Overview*

## ***Caretakers:***



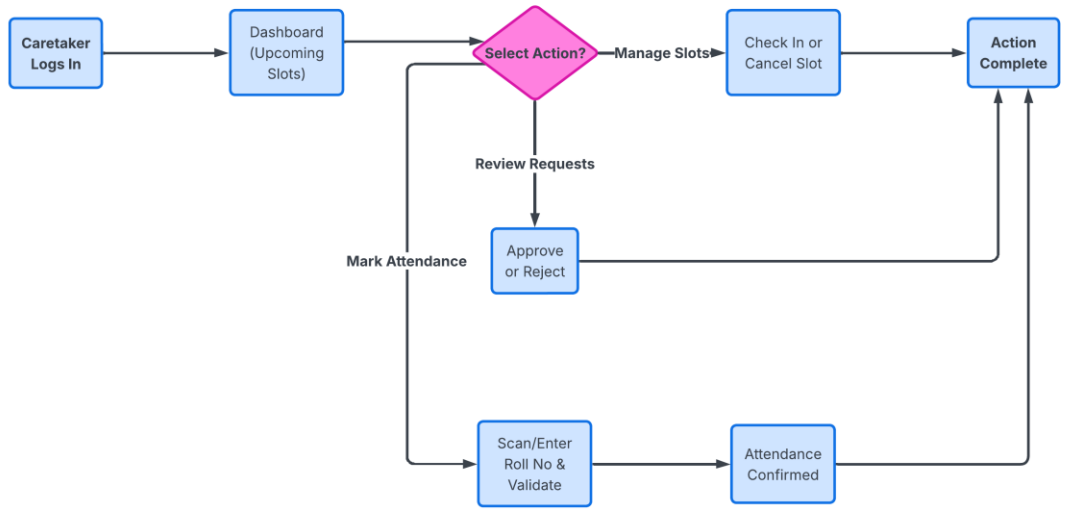
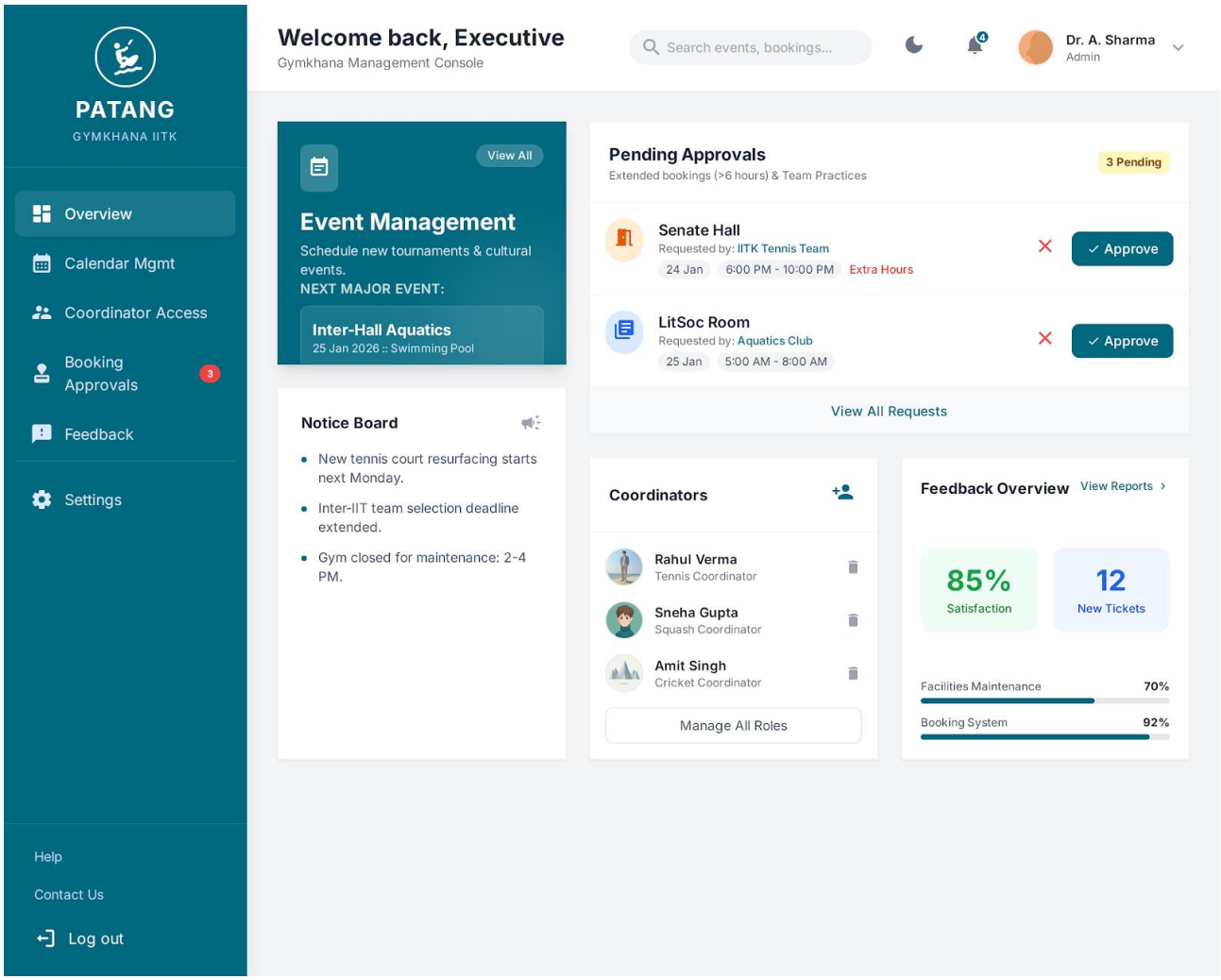




*Figure 2: Caretaker UI and Overview*

## ***Gymkhana Executive***



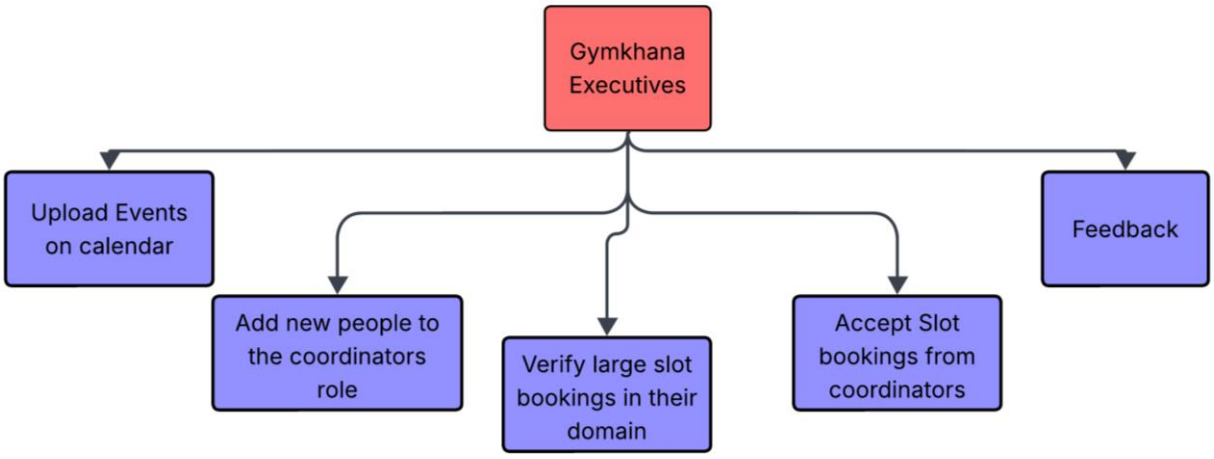




*Figure 3: Gymkhana Executive UI and Overview*

### 3.1.2 Hardware Interfaces

**Client Devices:**

- **Supported Device Types:**
  - Desktop Computers (Windows, macOS, Linux)
  - Laptops (Windows, macOS, Linux)
  - Smartphones and Tablets (iOS, Android)
- **Nature of Interaction:**
  - The website will send and receive data via standard internet protocols (e.g., HTTPS).
  - Responsive design ensures the website adapts to varying screen sizes and resolutions.

**Web Servers:**

- **Logical Characteristics:** The website will interface with the web server to host content and handle HTTP/HTTPS requests.
- **Physical Characteristics:** Compatible with servers running modern operating systems (Linux, Windows) equipped with sufficient CPU, RAM, and storage to handle expected traffic loads.

### 3.1.3 Software Interfaces

- The client-side components of the software must be able to operate without any issues with various modern web browsers like Apple Safari 10+, Google Chrome 44+, Microsoft Edge 90+, and Mozilla Firefox 40+.
- For ensuring authenticity of users the software shall verify credentials with the institute database using IITK Webmail Authentication (OTP-based).
- We will use databases for storing the following:
  - User login credentials and role-based access details (Students, Faculty, Gymkhana Executives, Caretakers).
  - Slot booking records and availability status for Sports Facilities (e.g., Badminton, Tennis), Gym and Swimming Pools.
  - Reservation logs for New-SAC shared facilities (e.g., Senate Hall, LitSoc Room).
  - Event details, categories, and registration links for the Gymkhana Calendar.
  - Feedback portal submissions and penalty records for slot misuse.
- MongoDB will be used as the database for backend.

## 3.2 Functional Requirements

### 3.2.1 F1: Sports Facilities Booking & Fair Use

- The platform shall provide a centralized digital system to manage and streamline the booking of sports facilities including Tennis, Badminton, Squash, Basketball, Table Tennis, and related amenities.
- To enforce equitable access and prevent hoarding, the system shall apply the following rules:
  - **Booking Window:** Users can book slots up to 3 days in advance.
  - **Quota System:** The system shall restrict each user to a maximum of two booking slots, totaling no more than two hours, within any rolling three-day window.
  - **Group Matching:** A "Look for Groups" feature will allow users to find others to play with, optimizing resource utilization by preventing solo usage of large courts. Such bookings cannot be made for the same day. If the required group is not filled by the booking day, the booking is automatically cancelled, and the slot is opened for normal booking.
  - **Penalty:** If a user fails to cancel a booking within a stipulated timeframe and does not show up, the caretaker marks the user absent, and the system applies penalties such as temporary suspension of booking privileges.

### 3.2.2 F2: Gym & Swimming Subscription Management

- The system shall automate the registration workflow for monthly, semesterly, or yearly memberships for the Gym and Swimming Pool, replacing manual paperwork.
- The workflow includes:
  - **Document Submission:** Users upload registration details and SBI-Collect payment receipts directly to the portal.
  - **Admin Verification:** Gym Administrators can view pending requests, verify payment validity, and approve or reject registrations.
  - **Digital Access Pass:** Upon approval, the system generates a unique, time-bound QR code linked to the user's ID/Roll Number. This QR code acts as a digital pass for entry and exit. Administrators will have a scanning interface to validate these QR codes in real-time to prevent unauthorized entry.

*Figure 4: Feature 2*

### 3.2.3 F3: New-SAC Resource Allocation

- Access to shared spaces like the Literary Society Room, Senate Hall, and Multi-purpose Room shall be restricted to Gymkhana Executives and Coordinators. Access rights can be delegated by the Gymkhana Executives to specific Club Coordinators via the admin dashboard.
- The booking system for these resources includes hierarchical approval logic:
  - **Standard Booking:** Approved club coordinators can book spaces for up to 6 hours per day and for up to 2 consecutive days, provided the room is available.

This consecutive-day limit is waived if the booking is made within 6 hours of the slot start time, subject to availability.
  - **Extended Use Escalation:** If a booking requires a duration exceeding the standard limits, the system will automatically flag the request and forward it to the respective General Secretary for manual approval.

### 3.2.4 F4: Operational Management & Overrides

- The system shall support special privileges for Coaches and Gymkhana Executives to accommodate team practices and institute events.
- Coaches and Gymkhana Executives can override standard booking rules to block specific time slots for team practices, maintenance, or institute events. These slots will appear unavailable to non-admin users.
- Caretakers at the facilities will have a dashboard to mark users absent if they fail to report on time. The system will immediately release these slots back to the pool, making them available for walk-in bookings to minimize resource wastage.

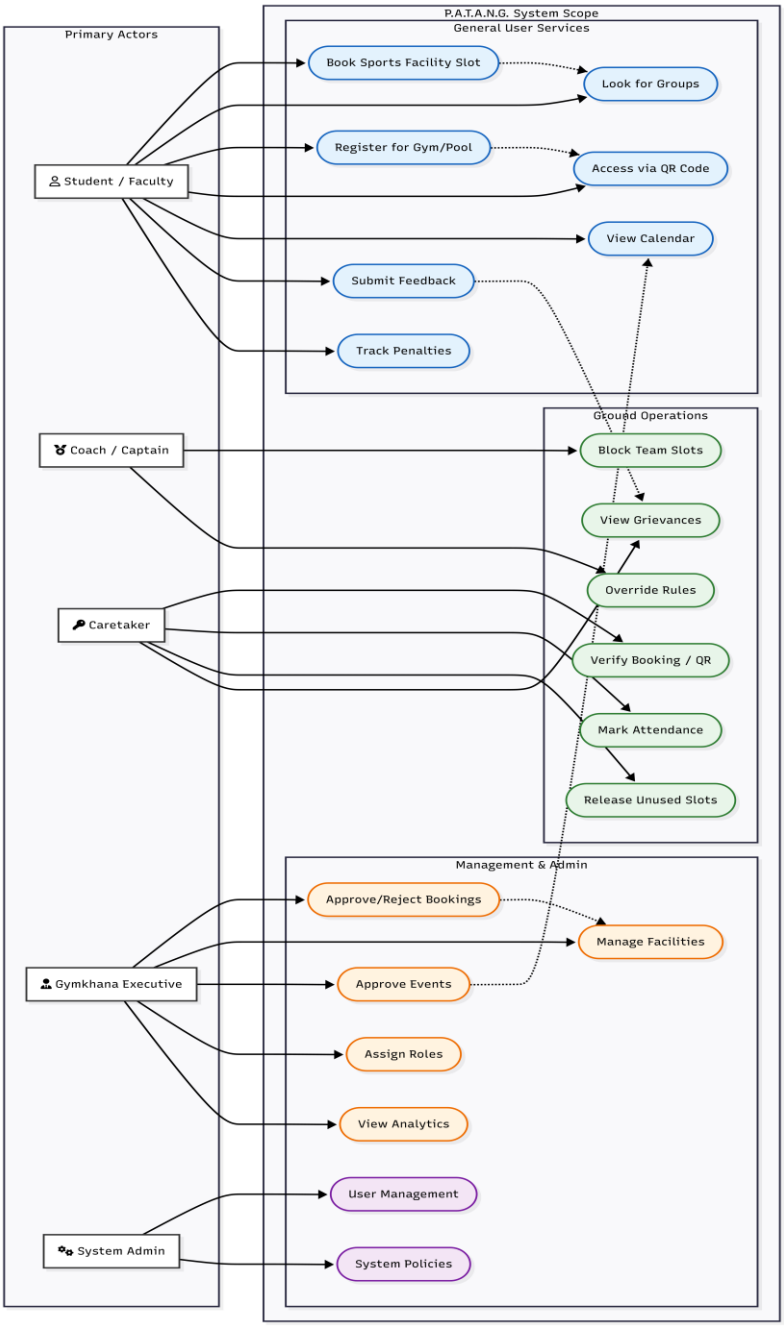
### 3.2.5 F5: Unified Gymkhana Calendar

- The platform will feature a categorized, theme-based calendar serving as the central hub for all campus activities.
- Coordinators and Activity Leaders can submit event details (Date, Time, Description, Registration Links, Eligibility) via a standardized form.
- Submitted events remain in a pending state until reviewed and approved by Executives.
- Once published, students can view events, filter them by category (e.g., Cultural, Technical, Sports), and navigate directly to registration portals.

### 3.2.6 F6: Feedback & Grievance Redressal

- The system will incorporate a unified feedback portal designed to handle two distinct streams of data:
  - **App Feedback:** Suggestions regarding the software interface or bugs, directed to the development team.
  - **Facility Feedback:** Grievances regarding equipment, cleanliness, or staff behavior at Gymkhana facilities.
- The system will route these inputs to the concerned authorities (e.g., a broken treadmill report goes to the Gym Admin; a login error goes to the IT team), ensuring that issues are rectified efficiently.

## 3.2 Use Case Model




### 3.2.1 Use Case #1 (Slot Booking)

**Author –** Aaryaman Aggarwal

**Purpose –** To enable students to book time slots for sports facilities (e.g. Tennis courts, Badminton courts) fairly and efficiently. The objective is to ensure optimal and fair utilization,

maintain booking discipline through ID verification, and enforce a penalty system for misuse or being absent.

**Requirements Traceability –** The requirements are as follows:

- The system shall allow only campus members to login through their IITK credentials to book slots for sporting facilities through the portal.
- They can book a maximum of **2 slots** (each slot is for an hour) over a **3-day window** to ensure fair use and not let a single person use it for multiple days
- Students could "Look for groups" if they do not have sufficient people to utilize the court. For example: A basketball court would need at least 6 people to play but if the user only has 2 people, he can open it up for other people to join as well.
- Students can indicate their interest to join such an opportunity to the system
- The system will only validate the booking if the slot is:
    - Free, i.e. not already reserved by someone else
    - Not blocked by coaches for Institute team practice
    - Not blocked by executives for event conditions
- On-ground staff can verify the booking using student/Faculty ID card/Roll number
- On-ground staff can mark the user as absent if the student doesn't check in within 10 minutes of the slot time.
- The system will penalize the users appropriately for no-shows unless it is cancelled within the allowed window (For example: At least 6 hours before)

**Priority**: High

**Preconditions**:

- Student must be logged in with a verified IITK email ID, verified through OTP
- Students cannot exceed the allowed quota of (2 slots over 3 days)
- The selected facility must be available (not already booked)
- Students must not be under a temporary penalty due to previous misuse.
- If the student doesn't meet the number of minimum students, he has the option to "Look for groups" to act as a matchmaking technique

**Post conditions**:

- The slot is successfully booked and reflected in the user's dashboard.
- Booking data is made available to caretakers/coaches.
- Student receives confirmation via email or on the dashboard
- Penalty logging if the student fails to show up.

**Actors** – Students, Faculty, Staff (Caretakers), Coaches

**Exceptions** – If a higher priority booking such as Institute team practice matches are scheduled then it may be such that the current booking is cancelled/reallocated depending on availability with appropriate notice. In this case, if the user is absent, **no penalty** will be applied.

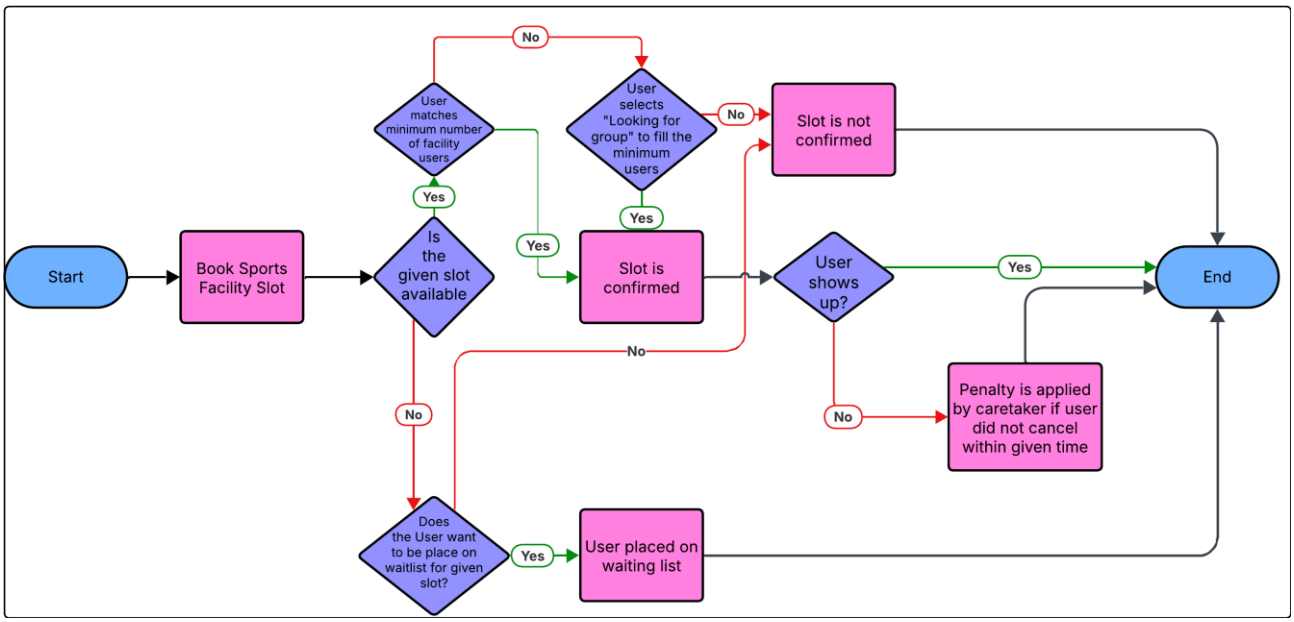



*Figure 5: Use case #1 overview*

### 3.2.2 Use Case #2 (New-Sac Bookings)

**Author** – Aryavart, Manas Todi

**Purpose** – To allow authorized coordinators to book New-SAC shared spaces efficiently while ensuring controlled access and hierarchical approval for extended usage.

**Requirements Traceability** –

- The system shall allow Gymkhana Executives to delegate booking rights to specific Club Coordinators.
- The system shall restrict booking access of New-SAC resources to Gymkhana Executives and approved Coordinators.
- The system shall allow approved coordinators to book resources for up to 6 hours per day and for a maximum of 2 consecutive days, subject to availability. This consecutive-day limit shall be waived if the booking is made within 6 hours of the slot start time.
- The system shall automatically escalate requests exceeding the standard duration to the respective General Secretary for approval.

**Priority** – High

**Preconditions** –

- The user must be assigned the role of Gymkhana Executive or approved Club Coordinator.

**Post conditions** –

- The user shall be able to successfully create a booking for New-SAC resources within permitted limits.
- The system shall consistently record approved bookings in the resource calendar.
- The General Secretary shall be able to view and act upon escalated booking requests.

- The user shall receive confirmation of booking status (approved, rejected, or pending approval).

**Actors** – Club Coordinators, General Secretary, System

**Exceptions** –

- If a requested slot is unavailable, the system shall prevent submission and prompt the user to choose a different slot.
- If escalation approval is rejected, the booking request shall be automatically cancelled.

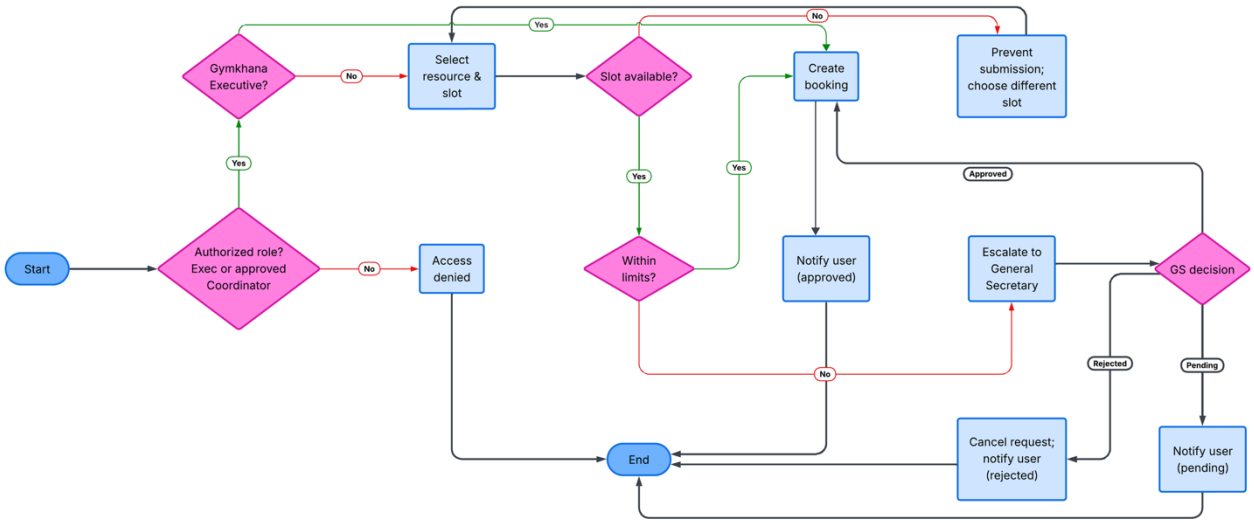



*Figure 6: Use case #2 overview*

### 3.2.3 Use Case #3 (Gym/Swimming Bookings)

**Author** - Aarin, Arsh

**Purpose** - To digitize the registration, payment verification, and daily access control for gym and swimming facilities.

**Requirements Traceability** -

- The system shall allow students and faculty to upload registration details and payment receipts (SBI Collect) for semesterly, monthly, or yearly subscriptions.
- The system shall allow Gym Administrators to review and approve/reject these registration requests.
- The system shall generate a unique, time-bound QR code (semesterly, monthly, or yearly) for approved users to grant access to the facilities.
- The system shall provide an interface for facility admins to scan user QR codes to verify validity and log entry/exit times.
- The system shall strictly enforce QR verification, preventing entry/exit if the scan occurs outside the user's designated time slot.
- The system shall allow coaches to block specific slots for institute team practices.

**Priority** - High

**Preconditions** -

- The user must be logged in with a valid institute email ID.
- The user must have completed the payment via SBI Collect and possess the transaction receipt.
- The Gym/Swimming administration must have defined the available subscription periods and slot capacities for the current term.

**Post conditions** -

- The user's profile is updated with a valid membership status (Active).
- A unique QR code is generated and accessible on the user's dashboard until the subscription ends.
- Access logs are updated in the database upon every successful scan at the facility entrance.
- Actors - Students/Faculty (End Users), Gym Administrator, Gatekeeper/Receptionist, System

**Exceptions** -

- Invalid Payment Receipt: If the administrator finds the uploaded receipt illegible or invalid, the request is rejected, and the user is notified to re-upload the correct document.
- Capacity Full: If the number of active registrations exceeds the facility's maximum capacity for the given period, the system prevents further new registrations until a slot opens.
- QR Code Expiry: If a user attempts to enter after their subscription period has ended, the scanner will flag the QR code as "Expired/Invalid" and deny entry.
- Overstay Penalty: If the entry/exit logs indicate a user has exceeded their allotted time slot significantly, the system flags the user for potential penalty or temporary suspension.

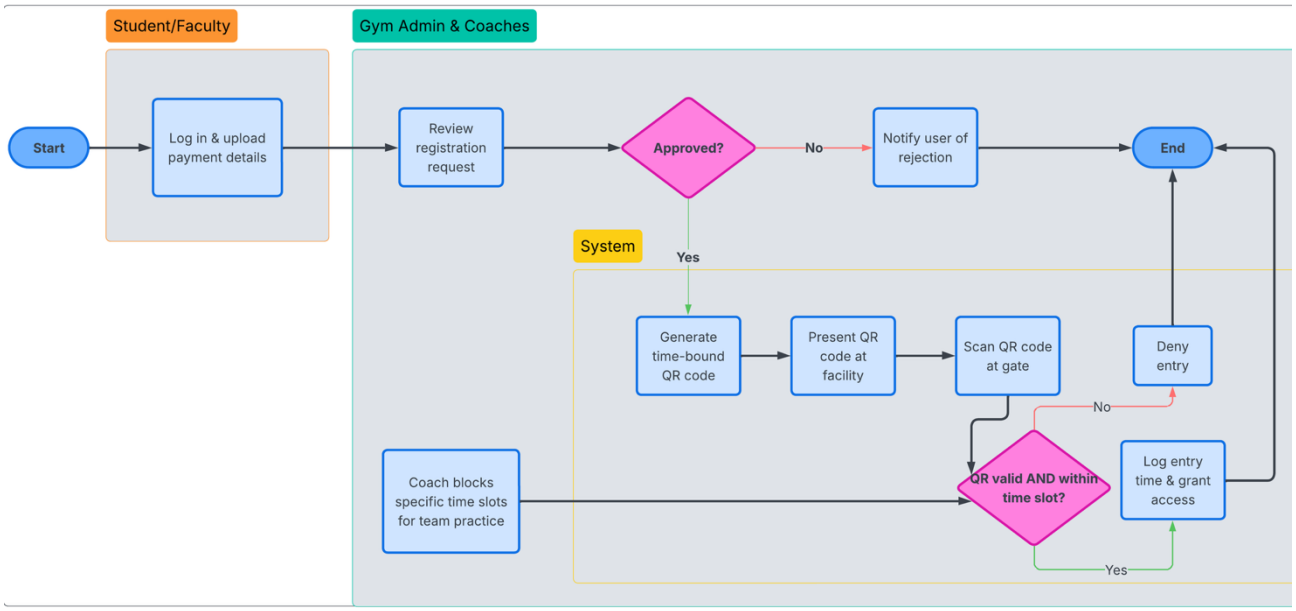



*Figure 7: Use Case #3 overviewFigure 8: Use Case #3 overview*

### 3.2.4 Use Case #4 (Gymkhana Calendar)

**Author** – Yash Bhardwaj, Shubh Bansal

**Purpose** – The primary objective of this use case is to provide a centralized, theme-based calendar for the Student Gymkhana that displays all ongoing and upcoming campus activities. The system streamlines event visibility for students while ensuring that events are reviewed and approved by higher executives before being published.

**Requirements Traceability** –

- **Event Creation and Submission** – Coordinators and club leaders must be able to submit event requests with complete details such as date, time, description, theme/category, registration links, and eligibility criteria.
- **Approval Workflow** – Higher executives must review submitted event requests and either approve, reject, or request modifications before the event appears on the calendar.
- **Calendar Display and Filtering** – Students must be able to view all approved events on a theme-based calendar and apply filters based on categories, dates, and eligibility.

**Priority** – Medium

**Pre Conditions** –

- Coordinators and leaders have the required permissions to submit event requests.
- Higher executives have approval rights.
- The system has an active calendar module with defined event categories.

**Post Conditions** –

- Approved events are published on the Gymkhana calendar.
- Rejected or pending events remain hidden from students.
- Students can view and filter events based on selected themes or categories.

**Actors** – Students, Event Coordinators / Club Leaders, Higher Executives.

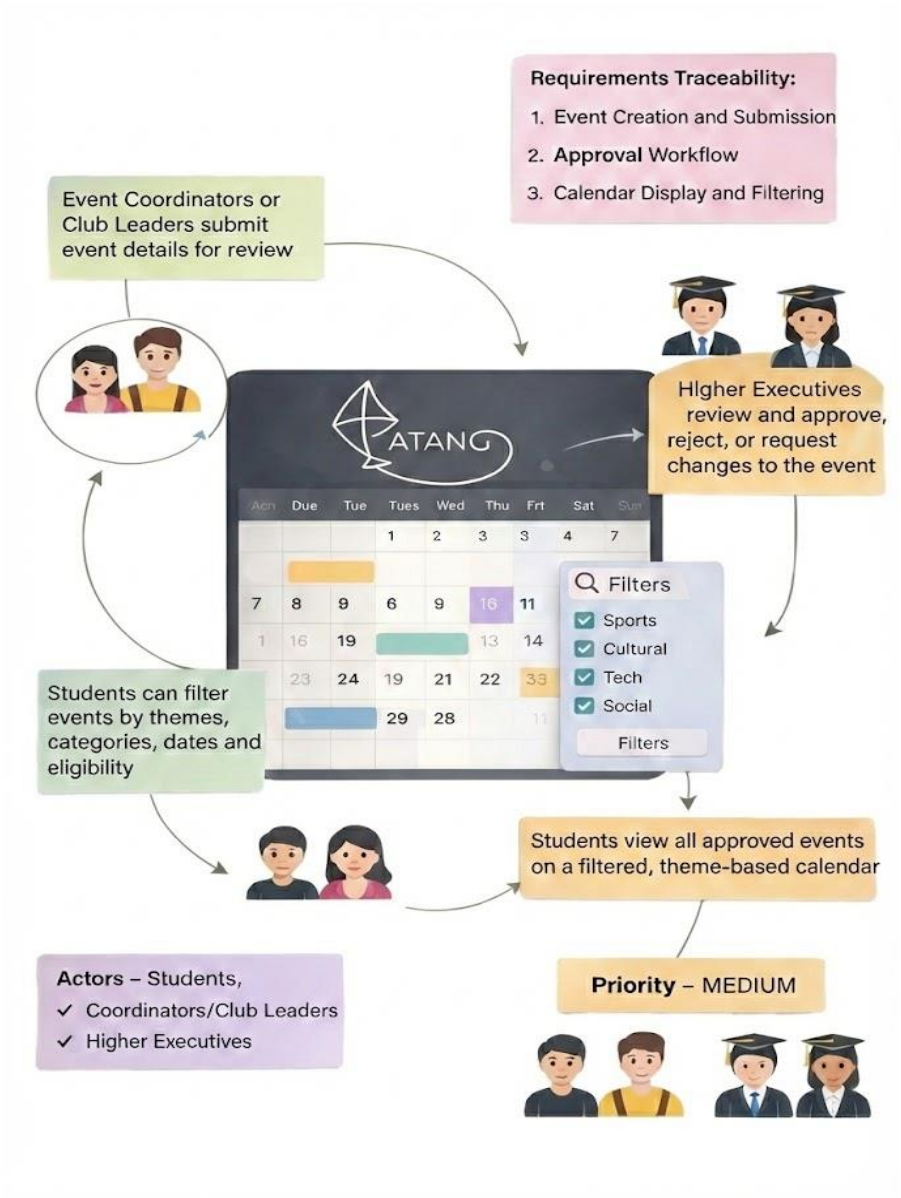



*Figure 8: Use Case #4 overview*

### 3.2.5 Use Case #5 (Feedback Portal)

**Author** – Sohom Chakraborty

**Purpose** – To provide a unified platform for students and faculty to report issues or suggest improvements regarding the P.A.T.A.N.G. software itself or specific campus facilities (e.g., broken equipment in Gym, unclean courts), ensuring the feedback reaches the relevant authorities for rectification.

**Requirements Traceability** –

- The system shall allow users to categorize their feedback into “Software Issues” (bugs, UI/UX suggestions) or “Facility Issues” (Gym, Sports Courts, New-SAC).
- The system shall route “Facility Issues” to the respective Gymkhana Executives or Caretakes responsible for that specific facility.
- The system shall route “Software Issues” to the development/maintenance team.
- The system shall allow administrators (Executives/Developers) to view received feedback and mark the status as “Resolved” or “In Progress” once addressed.
- The system shall allow users to track the status of the feedback they have submitted.

**Priority** – Medium

**Preconditions** –

- The user must be logged into the P.A.T.A.N.G. system using verified IITK credentials.
- The database of “concerned people” (Executives/Caretakes) linked to specific facilities must be up to date to ensure correct routing.

**Post Conditions** –

- The feedback is successfully stored in the database.
- A notification or email is sent to the relevant administrator (Executive or Developer) alerting them of a new issue.
- The user receives confirmation that their feedback has been recorded.

**Actors** – Students, Faculty, Gymkhana Executives, Caretakes, System

**Exceptions** –

- **Routing Failure:** If a specific facility does not have an active administrator assigned in the database, the feedback should default to the General Secretary or a Super Admin.
- **Database Timeout:** If the feedback fails to submit due to connectivity issues, the user must be notified to retry.

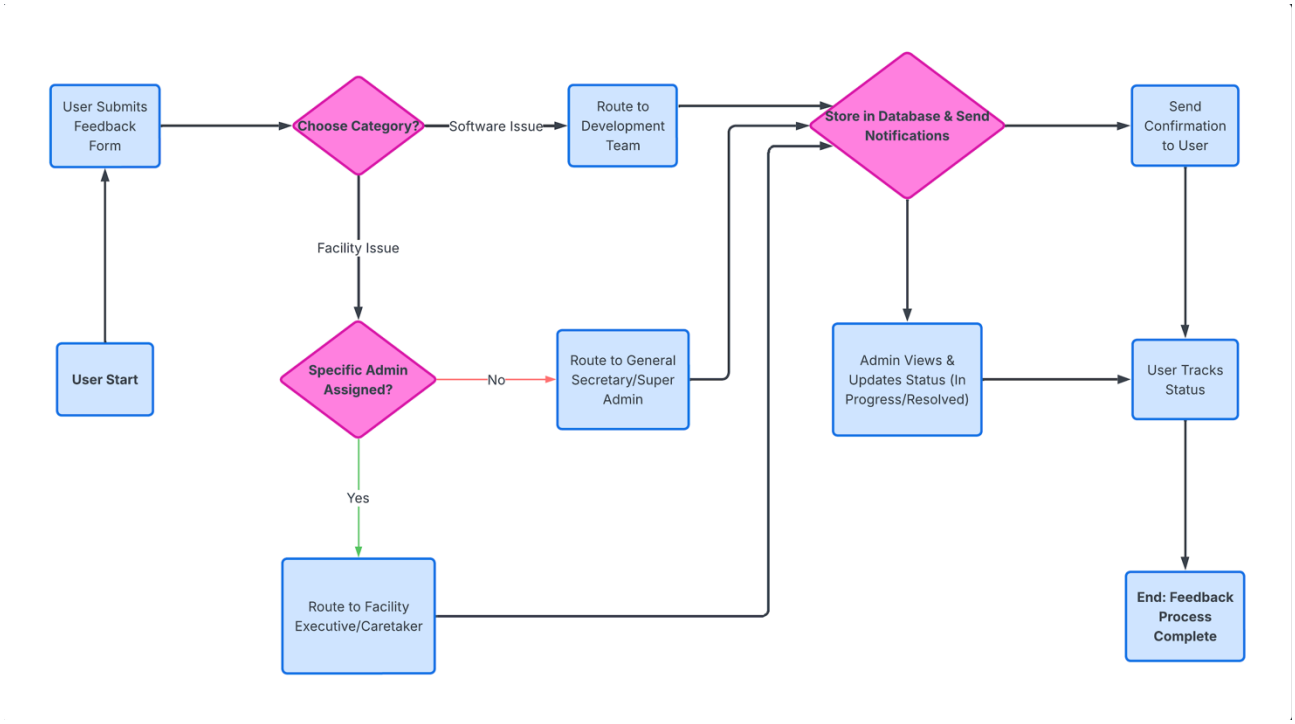



*Figure 9: Use Case #5 overview*

# 4 Other Non-functional Requirements

## 4.1 Performance Requirements

- **Concurrency** – The system shall be capable of supporting at least **1,500 concurrent users**, including students, faculty members, caretakers, coaches, club coordinators, and Gymkhana executives, during peak booking hours and major campus events.

- **Responsiveness** – For normal operations such as slot browsing, booking, QR verification, and calendar viewing, the average response time shall be **below 1 second**, and shall not exceed **3 seconds** for high-load operations such as payment verification uploads and bulk approval dashboards.

- **Compatibility** – The platform shall function correctly and maintain consistent layout and performance across modern web browsers (Chrome, Firefox, Safari, Edge) and mobile devices, including Android and iOS smartphones and tablets.

- **Availability** – The system shall maintain an operational availability of **99.9% over any 24-hour period**, with special emphasis on high-traffic windows such as daily slot-release times, gym registration periods, and major inter-hall or inter-college events.

- **Throughput** – The backend services shall be capable of processing at least **1,000 API requests per second** under peak load conditions, including booking queries, QR scans at entry gates, calendar fetches, and caretaker status updates.

## 4.2 Safety and Security Requirements

- **Login and Authentication Mechanism:**
  Users must log in using their IIT Kanpur email ID with OTP-based verification through the Institute mail system. Administrative roles such as Gymkhana executives and system admins will require additional verification for sensitive actions. All failed or suspicious login attempts shall be logged.

- **Role-Based Access Control (RBAC):**
  Access to features shall be strictly controlled based on the user's assigned role, such as student, caretaker, coach, or executive. Users will only be able to perform actions relevant to their responsibilities. Any attempt to bypass these restrictions will be recorded.

- **Secure Transmission of Information:**
  All communication between the client and server must be encrypted using HTTPS with TLS 1.3 or higher. This applies to booking requests, QR scans, payment uploads, and calendar updates. Unencrypted connections shall not be permitted.

- **Protection of Personally Identifiable Information (PII):**
  Sensitive details like roll numbers, photographs, and payment receipts must be encrypted while stored in the system. Only authorized staff such as administrators and finance officials will be able to view this data. Access to such records will be logged.

- **QR Code and Entry Security:**
  Entry to gyms and pools will be controlled using time-limited QR codes or ID-based verification tied to active registrations. QR codes will refresh frequently to prevent sharing or reuse. Every scan will be timestamped for auditing and penalty checks.

- **Penalty and Abuse Prevention Controls:**
  The system will automatically track no-shows, late cancellations, overstays, and misuse of slots. Penalties shall be applied according to predefined rules approved by the Gymkhana administration. Users will have the option to submit appeals for review.

- **Data Backup and Recovery:**
  The database must be backed up automatically at least once every 24 hours to prevent data loss. All backups will be encrypted and stored securely at multiple locations. Recovery procedures shall be tested periodically.

- **Payment and Receipt Security:**
  Uploaded SBI Collect receipts will be protected against tampering and linked to a single registration request. Every approval or rejection will be recorded with the official identity and timestamp. Duplicate or altered receipts shall be flagged.

- **Audit Logging and Monitoring:**
  All important actions such as slot blocking, role assignment, penalty overrides, and payment approvals shall be logged. These records will be accessible only to administrators and authorized institute officials. Logs must not be altered by regular users.

- **Browser and Device Security Requirements:**
  The system will support only modern browsers that follow current security standards. Access from outdated or insecure browsers may be restricted for safety reasons. Mobile devices must also meet basic security requirements.

- **Account Recovery and Access Revocation:**
  Account recovery will be handled through OTP verification sent to the registered IIT email address. Reset links or codes shall expire quickly to prevent misuse. User access will be revoked automatically when their institute's affiliation ends.

- **Compliance and Safety Policies:**
  The platform must follow IIT Kanpur IT usage policies and applicable data protection guidelines. Booking and access records will be retained only for administrative and safety needs. Any policy changes will be reflected in system updates.

## 4.3 Software Quality Attributes

### 4.3.1 Maintainability

- The system architecture, documentation, and codebase shall be modular so that new facilities, sports, booking policies, or user roles can be added with minimal disruption to existing functionality. Configuration driven rules will be used wherever possible to avoid hard-coding Institute policies.

- Fixing a critical bug or security issue, including updating documentation and running regression tests, should not take more than **two working days**, supported by automated testing pipelines and clear logging mechanisms.

- Adding a minor feature such as a new court type, additional slot duration, or notification template along with documentation and unit tests should require **less than one person-week** of development effort.

### 4.3.2 Availability

- In the event of a server failure or service outage, the platform must be restored to its most recent stable state within **two hours**, ensuring that bookings, access logs, and payment records are not lost.

- The system shall remain usable during high-traffic periods such as daily slot releases, gym registration windows, and major campus events, without noticeable degradation in basic functions like slot viewing or QR verification.

- Scheduled maintenance windows shall be announced in advance, and the system shall degrade gracefully by disabling non-critical features rather than blocking all users.

### 4.3.3 Reliability

- The Mean Time to Failure (MTTF) under normal campus usage shall exceed **one week**, including continuous operation during academic terms and sporting seasons.

- The platform shall undergo functional testing, stress testing for peak loads, and regression testing before every major release or semester rollout, especially booking logic, penalty rules, and access-control flows.

- The system must consistently display accurate information regarding slot availability, registration status, penalties, and calendar events, so that users can trust what they see when planning activities.

### 4.3.4 Portability and Interoperability

- The web application shall be responsive and fully functional across modern desktop and mobile browsers, allowing students and staff to access the system from laptops, phones, and tablets without layout or performance issues.

- Booking data, usage reports, and financial summaries shall be exportable in standard formats such as CSV or JSON to support administrative review and institutional reporting.

- Well-documented APIs shall be provided to allow future integration with Institute authentication systems, gate-scanner devices, payment portals, and other campus management software.

# Appendix A – Data Dictionary

## Data Dictionary

- **Common Class Variables**

| Name | Type | Description | States | Operations |
|---|---|---|---|---|
| name | string | Name of the User (Student/Faculty/Staff/ Executive | Any Text | Auto-filled from email or entered during registrations |
| email | string | Institute Email ID | Valid IITK email only | Used for login and authentication |
| pwd | string | Password for login | Alphanumeric | Specified during registration |
| role | string | Role of the user | Student, Faculty, Caretaker, Coach, Executive, Admin | Assigned on login or via role assignment |
| Roll_no | int | Roll Number of the user | Valid IITK Roll Number/Identification Number | Used for ID verification |

- **Common Class Methods**

| Method | Description |
|---|---|
| register() | To register a new user with IITK email |
| login() | Verify login via email and OTP |
| logout() | Logout of the current session |
| forgot_pwd() | Recover Password via OTP/Email |
| update_profile() | Update user information like phone number |
| view_bookings() | Fetch current and upcoming bookings |
| cancel_bookings() | Cancel a booking within permitted time to avoid penalty |

- **Facility Booking Variables**

| Name | Type | Description | States | Operations |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| facility_id | string | Unique ID for Sports Facility | e.g: Badminton_01 | Defined in master list |
| slot_time | datetime | Start time of slot | Future valid time | Selected by user during booking |
| booking_status | string | Booking status | Confirmed, Pending, Cancelled | Updated based on user or system action |
| group_required | bool | Is group formation required | True/False | Used for basketball |
| grup_filled | bool | Is group full? | True/False | If false, booking gets cancelled on D-day |

- **Facility Booking Methods**

| Method | Description |
|---|---|
| book_slot() | Book a slot for a facility under constraints |
| look_for_group() | Open the booking for group matchmaking |
| mark_absent() | Caretaker marks user absent if no show |
| release_slot() | System or caretaker releases unused slot for walk-ins |
| apply_penalty() | Penalize for misuse (e.g., no-show) |

- **Gym/Swimming Booking Variables**

| Name | Type | Description | States | Operations |
|---|---|---|---|---|
| subscription_type | string | Type of gym/swim plan | monthly / semester / yearly | Selected during registration |
| payment_receipt | string | Uploaded SBI Collect receipt | Valid URL or filename | Mandatory for approval |
| qr_code | string | Generated QR for access | Valid QR token | Used for gate entry/exit |
| access_valid_until | datetime | Expiry of QR/pass | Date | Controls access scope |

- **Gym/Swimming Booking Methods**

| Method | Description |
|---|---|
| upload_receipt() | Upload SBI Collect proof |
| approve_subscription() | Gym admin approves valid registration |
| generate_qr() | Create QR pass post approval |
| scan_qr() | Entry/exit verification at gate |

- **Feedback Portal Variables**

| Method | Description |
|---|---|
| submit_feedback() | Submit new feedback |
| update_status() | Admin marks progress |
| view_feedback() | User tracks submitted issue |

- **Gymkhana Calendar Variables**

| Name | Type | Description | States | Operations |
|---|---|---|---|---|
| event_id | string | Unique event identifier | Auto-generated | Used for internal reference |
| event_title | string | Name of event | Any text | Submitted by coordinator |
| event_category | string | Type of event | cultural, tech, sports, etc. | Used for filters |
| approval_status | string | Status of event | pending, approved, rejected | Controlled by executives |

- **Calendar Methods**

| Method | Description |
|---|---|
| submit_event() | Submit an event for calendar listing |
| approve_event() | Executive approves/rejects the event |

| | |
|---|---|
| filter_calendar() | Filter events by category/date |
| view_event_details() | Access full details and links |

- **Role Management Variables**

| Name | Type | Description | States | Operations |
|---|---|---|---|---|
| user_role | string | Role assigned to a user | student, coach, caretaker, etc. | Used for access control |
| access_rights | array | Allowed actions for role | List of permissions | Set by admin/executive |

- **Role Management Methods**

| Method | Description |
|---|---|
| assign_role() | Assign user role by admin/executive |
| revoke_access() | Remove a role or access |
| view_permissions() | View allowed actions per role |

## Data Relationships and Constraints

### Relationships

- Each user (student/faculty) can make bookings for sports or subscriptions.
- Facility slots are tied to both time and group constraints.
- Admins, Caretakers, and Executives have different oversight rights.
- Events and feedback are routed to roles based on type and facility mapping.

### Constraints

- No user can book >2 slots in a 3-day window.
- Bookings require valid login via verified email.
- Group-required sports must meet player threshold.
- SBI Collect receipt must be valid (no duplicates).
- QR access must be within time bounds.

# Appendix B - Group Log

● A WhatsApp group was created for communication between members.

● Regular offline meetings were hosted whenever necessary to take inputs from members on the project progress.

| Date and Time of the Meet | Meet Type | Ongoing Discussions in the Meet |
|---|---|---|
| 17/01/2026<br>9:00 am –12:00 am | Offline Meet | Meet to discuss the initial project idea and elaborate on real-world relevance. |
| 18/01/2026<br>9:00 am –12:00 am | Offline Meet | Discussion over what to include in software functionality. Initial ideas on the requirement document were shared. |
| 20/01/2026<br>9:00 pm – 2:00 am | Offline Meet | Further discussion on use cases and related features. Tasks associated with the initial draft (SRS doc) were equally distributed to each team member. |
| 23/01/2026 8:00 pm | Offline Meet | First draft was reviewed by assigned TA and changes were suggested |
| 23/01/2026<br>9:00 pm – 1:00 am | Offline Meet | A meeting was held to make changes to the initial draft, ensuring that each team member contributed equally to its completion. |
| 23/01/2026<br>9:00 pm – 1:00 am | Offline Meet | A meeting was held to work on the UI for the draft. |
| 24/01/2026<br>1:00 pm – 4:00 pm | Offline Meet | A meeting was held to prepare the final draft, ensuring that each team member contributed equally to its completion. |